Contents

Coordination Physics Research Agenda

A Program for Validating and Extending the Framework

From Theory to Empirical Science

Executive Summary

The 17 Laws of Civilizational Coordination provide a theoretical framework that makes specific, testable predictions. This research agenda outlines the experiments, observations, and analyses needed to validate, refine, and extend this framework.

Core Questions: 1. Is $\theta \approx 0.375$ truly universal, or does it vary with context? 2. Can we measure H_3 in real-time with sufficient precision? 3. Does AI actually accelerate coordination dynamics as predicted? 4. Can we identify early warning signals before threshold crossing?

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1. Immediate Research Priorities (2025-2027)

1.1 H_3 Measurement Development

Objective: Develop reliable, real-time, cross-cultural trust measurement

Research Questions: - What combination of surveys, behavioral measures, and passive sensing provides the most accurate H_3 estimate? - How quickly can H_3 changes be detected? - What is the measurement error, and how does it affect threshold predictions?

Proposed Methods:

Method	Data Source	Frequency	Validation
Survey fusion	WVS, Gallup, Edelman	Quarterly	Cross-validation
Behavioral proxies	Transaction data, cooperation metrics	Daily	Survey correlation
Passive sensing	Social media, search trends	Real-time	Behavioral correlation
Network analysis	Communication patterns	Weekly	Trust game correlation

Deliverables: 1. Standardized H_3 measurement protocol 2. Open-source measurement toolkit 3. Validation study across 50+ countries 4. Real-time H_3 dashboard (prototype)

Funding Need: $3-5M over 2 years

1.2 Threshold Validation

Objective: Empirically confirm $\theta \approx 0.375$ and characterize its precision

Research Questions: - Does the threshold vary by context (national, organizational, community)? - What is the uncertainty range on θ? - Are there secondary thresholds for different coordination modes?

Proposed Methods:

1. **Historical analysis extension**
    - Expand from 48 to 100+ historical cases
    - Include organizational and community collapses
    - Improve H_3 reconstruction methodology
2. **Experimental economics**
    - Public goods games with varying initial trust
    - Identify behavioral threshold for cooperation collapse
    - Cross-cultural replication
3. **Agent-based modeling**
    - Simulate coordination dynamics across parameter space
    - Identify phase transition boundaries
    - Test robustness to model assumptions
4. **Natural experiments**
    - Identify contemporary cases near θ
    - Track trajectory and outcome
    - Compare predictions to observations

Deliverables: 1. Refined θ estimate with confidence intervals 2. Context-dependency analysis 3. Validated simulation model 4. Prospective prediction study design

Funding Need: $5-8M over 3 years

1.3 AI Coordination Impact Assessment

Objective: Quantify A_positive and A_negative in real deployments

Research Questions: - Which AI applications have the largest coordination impact? - Can we predict coordination effects before deployment? - What design features distinguish positive from negative AI?

Proposed Methods:

1. **Platform analysis**
    - Measure polarization changes after algorithm modifications
    - Track cross-group exposure metrics
    - Correlate with trust survey changes
2. **Controlled deployment studies**
    - A/B tests of AI designs on coordination outcomes
    - Measure before/after H_3 in treatment populations
    - Identify causal mechanisms
3. **LLM behavioral analysis**
    - Test AI systems for polarization amplification
    - Measure perspective balance in responses
    - Develop coordination alignment benchmarks
4. **Economic impact tracking**
    - Monitor AI-driven displacement effects
    - Correlate with regional trust changes
    - Model feedback loops

Deliverables: 1. AI Coordination Impact Database 2. Pre-deployment assessment methodology 3. Design guidelines for coordination-positive AI 4. Policy recommendations with evidence base

Funding Need: $10-15M over 3 years

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2. Medium-Term Research (2027-2030)

2.1 Early Warning System Development

Objective: Build operational early warning for coordination crises

Research Components:

1. **Indicator development**
    - Variance increase before threshold
    - Critical slowing down detection
    - Autocorrelation monitoring
    - Polarization trajectory analysis
2. **Prediction models**
    - Machine learning on historical patterns
    - Real-time trajectory forecasting
    - Confidence calibration
3. **Alert system design**
    - Threshold triggers for intervention
    - Communication protocols
    - False positive management

Deliverables: 1. Operational early warning system (10+ countries) 2. False positive/negative rate assessment 3. Intervention trigger protocol 4. Integration with policy processes

Funding Need: $15-20M over 3 years

2.2 Intervention Effectiveness Research

Objective: Identify which interventions actually raise H_3

Research Questions: - Which trust-building interventions have the largest effect size? - How long do intervention effects persist? - What conditions determine intervention success?

Proposed Methods:

1. **Meta-analysis**
    - Comprehensive review of trust-building interventions
    - Effect size standardization
    - Moderator analysis
2. **Randomized controlled trials**
    - Community-level trust interventions

  - Platform design modifications
  - Economic redistribution experiments
3. **Policy evaluation**
   - Natural experiments from policy changes
   - Difference-in-differences analysis
   - Long-term follow-up

Deliverables: 1. Intervention effectiveness hierarchy 2. Cost-effectiveness estimates 3. Implementation guidelines 4. Scaling recommendations

Funding Need: $20-30M over 4 years

2.3 Network Topology and Redundancy

Objective: Map coordination infrastructure and identify vulnerabilities

Research Components:

1. **Network mapping**
   - Identify coordination pathways (economic, social, political)
   - Measure redundancy (R) in real networks
   - Model cascade propagation
2. **Vulnerability analysis**
   - Identify critical nodes
   - Simulate failure scenarios
   - Quantify fragility
3. **Redundancy building**
   - Design more robust architectures
   - Test interventions to increase R
   - Monitor resilience changes

Deliverables: 1. Global coordination network map 2. Vulnerability assessment methodology 3. Redundancy building playbook 4. Resilience monitoring system

Funding Need: $10-15M over 3 years

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3. Long-Term Research (2030+)

3.1 Metacognitive Capacity Assessment

Objective: Measure and enhance civilizational self-awareness

Research Questions: - How do we measure M (metacognitive capacity)? - What determines the gap between M and C_env? - Can we increase M faster than C_env grows?

Research Components:

1. **M measurement**
   - Collective prediction accuracy
   - Response time to crises

- Model quality assessment
2. **M enhancement**
   - Collective intelligence platforms
   - Decision support systems
   - Institutional learning mechanisms
3. **C_env tracking**
   - Environmental complexity monitoring
   - Rate of change measurement
   - Predictability analysis

3.2 Cross-Civilizational Studies

Objective: Prepare for eventual contact with other civilizations

Research Components:

1. **SETI coordination**
   - Develop protocols for H_3 assessment of detected signals
   - Plan for contact implications on Earth coordination
   - Design communication for post-threshold civilizations
2. **Universal coordination physics**
   - Test whether θ is truly universal
   - Model coordination in non-human systems
   - Explore fundamental limits

3.3 Quantum Limits on Coordination

Objective: Explore fundamental physical constraints on coordination

Research Questions: - Are there quantum limits on coordination at large scales? - How does light-speed communication affect θ? - What is K_max for a Type II+ civilization?

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4. Methodological Innovations Needed

4.1 Measurement Innovations

Innovation	Current Status	Development Need
Real-time H_3	Not available	High priority
Passive trust sensing	Early research	Medium priority
Cross-cultural calibration	Limited	High priority
Dark trust measurement	Theoretical	High priority
Network-based H_3	Conceptual	Medium priority

4.2 Analytical Innovations

Innovation	Current Status	Development Need
Threshold detection algorithms	Early stage	High priority
Cascade prediction models	Theoretical	High priority
Intervention optimization	Limited	Medium priority
Multi-scale coordination models	Conceptual	Medium priority

4.3 Computational Innovations

Innovation	Current Status	Development Need
Agent-based coordination models	Basic	Medium priority
Real-time monitoring systems	Not available	High priority
AI coordination impact simulation	Limited	High priority
Early warning platforms	Not available	High priority

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5. Institutional Infrastructure

5.1 Research Centers

Proposed: International Coordination Physics Institute

- **Location**: Multiple nodes (Europe, Americas, Asia)
- **Mission**: Advance coordination science for civilizational benefit
- **Functions**:
  - Core research on coordination dynamics
  - H_3 monitoring infrastructure
  - Early warning system operation
  - Policy translation and advisory
  - Training and capacity building

Estimated annual budget: $50-100M

5.2 Data Infrastructure

Proposed: Global Coordination Observatory

- **Function**: Continuous H_3 monitoring across all countries
- **Data**: Survey integration, behavioral proxies, network analysis
- **Output**: Real-time H_3 estimates, early warning alerts, research data
- **Governance**: Multi-stakeholder, independent, transparent

Estimated annual budget: $20-30M

5.3 Policy Interface

Proposed: Coordination Impact Assessment Authority

- **Function**: Evaluate technology and policy for coordination impact
- **Authority**: Recommend (initially), possibly mandate (later)
- **Scope**: AI systems, platform changes, major policies
- **Model**: Similar to environmental impact assessment

Estimated annual budget: $10-20M

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6. Collaboration Framework

6.1 Disciplinary Integration

Discipline	Contribution	Integration Point
Physics	Phase transition theory	Laws 13-14
Economics	Game theory, mechanism design	Laws 1, 7, 16
Psychology	Trust measurement, behavior	Laws 6, 8, 9
Sociology	Network analysis, institutions	Laws 3, 10
Computer Science	AI systems, monitoring	Law 17
Political Science	Governance, policy	Laws 7, 11
History	Case studies, validation	All laws
Philosophy	Ethics, metacognition	Laws 15, 17

6.2 Key Partnerships

- **Academic**: Complex systems institutes, trust research centers
- **Government**: National security, science agencies
- **Tech**: AI labs, platform companies
- **International**: UN agencies, OECD, G20

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7. Funding Strategy

7.1 Funding Sources

Source	Amount	Timeline	Purpose
Government grants	$50-100M	2025-2030	Core research
Foundation funding	$20-50M	2025-2030	High-risk innovation
Tech company investment	$30-50M	2025-2030	Applied research
International orgs	$20-30M	2025-2030	Global coordination
Philanthropic	$50-100M	2025-2030	Institution building

7.2 Total Investment Needed

Phase 1 (2025-2027): $20-30M/year = $60-90M total **Phase 2 (2027-2030)**: $50-80M/year = $150-240M total **Phase 3 (2030+)**: $100-150M/year ongoing

Total 2025-2035: ~$1-2B

This is approximately: - 0.01% of global AI investment - 0.001% of global defense spending - An order of magnitude less than the cost of one coordination crisis

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8. Success Metrics

8.1 Research Metrics

Metric	Target 2027	Target 2030
H_3 measurement accuracy	±0.05	±0.02
Threshold precision	θ ± 0.02	θ ± 0.01
Early warning lead time	6 months	12 months
Intervention effect sizes known	10 interventions	50 interventions
Countries with real-time H_3	20	100

8.2 Impact Metrics

Metric	Target 2027	Target 2030
Policy adoption	5 countries	20 countries
AI assessment implemented	3 companies	20 companies
Early warnings issued	2	10
Crises averted (estimated)	1	5
H_3 trend (global)	Stable	Rising

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9. Risks and Mitigation

9.1 Research Risks

Risk	Probability	Mitigation
θ varies more than expected	Medium	Develop context-specific models
H_3 measurement proves unreliable	Low	Multiple independent methods

Risk	Probability	Mitigation
AI effects not measurable	Low	Focus on natural experiments
Framework fundamentally wrong	Low	Design falsifiable tests early

9.2 Implementation Risks

Risk	Probability	Mitigation
Insufficient funding	Medium	Diversify sources, demonstrate value
Political resistance	High	Frame as non-partisan, focus on evidence
Misuse of early warning	Medium	Careful governance, transparency
Tech company non-cooperation	High	Regulatory pathway, public pressure

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10. The Research Imperative

10.1 Why This Research Matters

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Current situation:
- H₃ declining globally
- AI accelerating dynamics
- No validated early warning system
- Interventions not evidence-based
- Window for action: 1-3 years

Without this research:
- We cannot predict crises
- We cannot identify effective interventions
- We cannot measure progress
- We fly blind into the critical window

With this research:
- Early warning becomes possible
- Interventions can be optimized
- Progress can be measured
- Evidence can guide policy
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10.2 The Ultimate Question

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This research program asks:

Can humanity develop the science of coordination
fast enough to survive the technology we've already created?

The answer is not yet determined.

The research outlined here is our attempt to tip the balance
toward survival.
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"Science is humanity's way of understanding itself. Coordination physics is the science of our collective survival. This research agenda is our attempt to learn fast enough to live."

This research agenda accompanies Paper 2C: The 17 Laws of Civilizational Coordination